Anthropometric and Hematological Variables in Male Wistar Rats Fed with High Fructose Drink and Vitamin C

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ABSTRACT

There is a global increase in fructose intake through sweetened beverages and high fructose consumption has been linked to metabolic disturbances. This study aimed to investigate the impact of vitamin C supplementation on anthropometric and hematological variables in rats fed a high-fructose drink. Twenty male Wistar rats were randomly divided into four groups; group 1 (control), group two treated vitamin C (1g/100ml), group three treated high-fructose drink (HFD) (15g/100ml), and group four treated high-fructose drink + vitamin C. At the end of the 28th day, anthropometric variables were measured, and blood samples were collected for analysis of hematological variables and indices. Statistical analysis was performed using one-way ANOVA with GraphPad at $p \leq 0.05$. The results showed a significant increase in anthropometric variables for the group fed HFD. This significant increase was observed in body weight (221.3 ± 4.06g vs 270.7 ± 12.17g), abdominal circumference (6.17 ± 0.33 vs 7.5 ± 0.29 cm), and BMI (3.06 ± 0.06 vs 3.94 ± 0.24 g/cm^2) compared to control. On the other hand, HFD non-significantly decreased red blood cell count, white blood cell count, platelet count, mean corpuscular hemoglobin and mean corpuscular hemoglobin concentration while treatment with vitamin C improved these hematological parameters. This study therefore established that HFD may downregulate hematological parameters through increased body weight and abdominal circumference via lipogenesis and adiposity.

Keywords: Anthropometric, Hematological Variables, Fructose, Ascorbic Acid.

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1. INTRODUCTION

The global increase in the consumption of fructose, primarily through sweetened beverages and processed foods, has raised significant concerns about its potential health implications. Fructose, a monosaccharide commonly found in fruits and honey, is widely used in the food industry due to its high sweetness index and low production cost. However, excessive dietary intake of fructose has been linked to various metabolic disorders, including obesity, type 2 diabetes, non-alcoholic fatty liver disease (NAFLD), and cardiovascular diseases [1; 2]. As the prevalence of these diseases rises, understanding the biological impact of high fructose consumption is becoming increasingly important. Hematological parameters are critical indicators of overall health and can reflect alterations in immune function, oxygen transport, and inflammation. Changes in these variables due to high fructose intake could have significant implications for metabolic health.

Vitamin C (ascorbic acid) is a well-known antioxidant that plays a critical role in neutralizing free radicals and reducing oxidative stress [3]. It is also involved in numerous physiological processes, including collagen synthesis, immune function, and iron absorption [4]. Due to its antioxidant properties, vitamin C may counteract the adverse effects of fructose-induced oxidative stress and inflammation. Several studies have

suggested that vitamin C supplementation can mitigate metabolic disturbances caused by high fructose consumption [5]. The rising consumption of fructose-rich diets poses a significant public health concern due to their association with metabolic diseases. While the detrimental effects of high fructose consumption on body weight, organ weight, and metabolic health are well documented [6], there is a gap in understanding how such diets specifically affect hematological variables and anthropometric variables. Additionally, the potential role of vitamin C in mitigating these effects is not well understood. However, the specific effects of vitamin C on anthropometric and hematological variables in the context of high fructose intake remain unclear.

Given the potential interaction between high fructose consumption and vitamin C supplementation, this study aims to explore the effects of both substances on anthropometric and hematological variables in male Wistar rats. By using an established animal model, this research seeks to provide valuable insights into the physiological impact of a high-fructose diet and the possible protective role of vitamin C. Furthermore, while vitamin C has been studied for its antioxidant properties, its direct influence on hematological parameters and body composition in the context of high fructose consumption requires further investigation.

Materials and Methods

Experimental design: Twenty adult male Wistar rats of comparable weights (120 ± 20g) were used for this study. The rats were obtained from the animal house, Department of Physiology, College of Medicine, Ambrose Alli University, Ekpoma and transferred to the Department of Physiology animal holding facility in the same institution. They were housed in plastic cage with a wire screen top. The animals were fed with drinking water and pelleted rat feed during acclimatization (14 days) and were maintained under standard care of laboratory animals as stated in the guidelines and regulations for the use of animals in scientific experiment.

Substance preparation and administration: Vitamin C tablets and Fructose were obtained from a standard pharmacy in Ekpoma, Edo state. The Vitamin C tablets were crushed into powder using a laboratory mortar and pestle. The high fructose drink was prepared by dissolving 30g of fructose in 100ml of distilled water, the vitamin C drink was prepared by dissolving 1g of vitamin C in 100ml of distilled water while the high fructose and vitamin C drink mixture was prepared by adding 30g of fructose and 1g of vitamin C to 100ml of distilled water. The correct quantities of Fructose and Vitamin C was properly measured using an electronic sensitive scale, the distilled water was measured using a measuring beaker and the different water compositions were prepared daily for the duration of the experiment.

Experimental grouping: After acclimatization, the rats were assigned to four groups of five rats each as follows: Group 1 (Control group) were given normal rat feed and 100ml of distilled water. Group 2 (Vit C): were given normal rat feed and 100ml of distilled water with 1g vitamin C. Group 3 (HFD): were given normal rat feed and 100ml of distilled water with 30g fructose. Group 4 (HFD+Vit C): were given normal rat feed and 100ml of distilled water with 1g vitamin C and 30g fructose. The administration lasted for 28 days.

Sample collection: At the end of the experiment, blood samples were obtained following standard laboratory procedures by means of cardiac puncture with a needle and syringe and collected into EDTA bottles for analysis.

Sample analysis: Body weight (using electronic weighing scale), body length and Abdominal Circumference (using measuring tape) were determined before the study

commences, after every seven days and at the end of the study to assess changes in the anthropometric variables. The blood in EDTA bottle was used for the analysis of hematological variables and hematological indices. Full Automatic Blood Cell Counter (PCE-210N Ver. 5.42N) was used for assaying the different blood components.

Statistical analysis: Data were analyzed with GraphPad Prism version 8.0 (GraphPad Software, San Diego, CA) and expressed as the means ± SEM. One-way ANOVA was used for comparisons, followed by post hoc Newman-Keuls Multiple Comparison test. Results were statistically significant at $P \leq 0.05$.

RESULTS

Anthropometric variables: Table 1 shows the effect of HFD and Vitamin C on anthropometric variables in male Wistar rat. There was a significant increase in body weight (221.3±4.06g vs 270±12.17g), abdominal circumference (6.17 ± 0.33 vs 7.50 ± 0.29 cm) and body mass index (3.06 ± 0.06 vs 3.94 ± 0.24 g/cm^2) in the HFD compared to the control group. However, body length was not significantly different between groups.

Hematological variables: Table 2 shows the effect of HFD and Vitamin C on hematological variables in male Wistar rat. Red blood cell count, White blood cell count and platelet count were not significantly different between groups, they were however, lower in the HFD group compared to the control and vitamin C treated groups.

Hematological indices: Table 3 shows the effect of HFD and Vitamin C on hematological indices in male Wistar rat. Hematological indices herein estimated were not significantly different between groups, however, pack cell volume, hemoglobin and mean corpuscular volume were higher while mean corpuscular hemoglobin and mean corpuscular hemoglobin concentration were lower in the HFD compared with the control.

Table 1. The anthropometric variables

Group	Body weight (g)	Body length (cm)	Abdominal circumference (cm)	BMI (g/cm^2)
CN	221.3 ± 4.06	8.5 ± 0.00	6.17 ± 0.33	3.06 ± 0.06
Vit. C	208.7 ± 12.33	7.73 ± 0.27	6.50 ± 0.00	3.14 ± 0.08
HFD	270.7 ± 12.17*	8.3 ± 0.10	7.50 ± 0.29*	3.94 ± 0.24*
HFD + Vit C	248.7 ± 8.11	8.3 ± 0.10	7.00 ± 0.00	3.61 ± 0.14

Values are mean ± SEM, * significant vs CN. CN = Control group, Vit C = Vitamin C, HFD = High Fructose Drink, HFD+Vit C = High Fructose Drink and Vitamin C, BMI = Body mass index.

Table 2 shows Hematological variables

GROUP	RBC	WBC	Platelets
CN	7.53 ± 0.15	6.10 ± 0.55	697.00 ± 48.79
Vit c	7.99 ± 0.21	7.80 ± 0.17	690.30 ± 25.46
HFD	6.95 ± 0.24	5.50 ± 0.45	600.00 ± 39.40
HFD + Vit C	7.26 ± 0.64	6.23 ± 1.65	690.00 ± 40.67

Values are mean ± SEM. CN = Control group, Vit C = Vitamin C, HFD = High Fructose Drink, HFD+Vit C = High Fructose Drink and Vitamin C, RBC = Red blood cell, WBC = White blood cell

Table 3 shows Hematological Indices

GROUP	Pack Cell Volume (PCV) (%)	Hemoglobin (Hg) (g/dm)	Mean Corpuscular volume (MCV)	Mean Corpuscular Hemoglobin (MCH) (Pg)	Mean Corpuscular Hemoglobin Concentration (MCHC)
CN	33.27 ± 5.58	11.30 ± 1.77	50.90 ± 0.78	36.33 ± 18.68	71.53 ± 38.51
Vit C	42.40 ± 1.22	14.07 ± 0.34	53.10 ± 0.82	17.67 ± 0.26	33.20 ± 0.20
HFD	36.73 ± 1.50	13.57 ± 1.47	52.80 ± 0.45	18.20 ± 0.20	34.23 ± 0.12
HFD+ Vit C	38.53 ± 3.13	13.27 ± 0.94	53.10 ± 0.42	18.30 ± 0.35	33.47 ± 0.92

Values are mean ± SEM, CN = Control group, Vit C = Vitamin C, HFD = High Fructose Drink, HFD+Vit C = High Fructose Drink and Vitamin C.

4. DISCUSSION

This study showed that HFD significantly increase body weight, abdominal circumference and BMI. This is in line with the Mamikutty *et al.* [7], who reported an increased body weight in Wistar rats fed with HFD. The mechanism behind this reported increase is linked to the fact that fructose is known to promote lipogenesis and in turn lead to increase in the body weight. The metabolism of fructose generates acetyl CoA, which is a main building block for the synthesis of fatty acid which contributes greatly to lipogenesis [1].

Some studies have reported that Vitamin C has a protective effect on body weight when it is administered alongside a high-fructose diet [8, 9]. In fact, studies have reported that male Wistar rats fed with a high-fructose diet with concurrent vitamin C supplementation had significantly lower weight gain [8, 9]; a finding also observed in this study. Similarly, Johnson et al. [10] found that male Wistar rats consuming high levels of fructose developed insulin resistance, directly associated with increase in body weight and adiposity. These studies suggest that fructose-induced weight gain may be mediated through mechanisms involving insulin resistance and altered lipid metabolism. The authors suggested that vitamin C's antioxidant properties might mitigate oxidative stress, thereby reducing fat accumulation by limiting lipid synthesis which inhibits weight gain [8-10]. The finding of this study correlates with those that are established in literatures.

The increase in abdominal circumference observed in the HFD group indicates enhanced visceral fat deposition, a common consequence of high fructose intake. Farah et al. [8] opined that increased abdominal circumference has a high clinical relevance and a screening assessment in the determination of central obesity. However, the vitamin C treated groups showed a significant decrease in abdominal circumference when compared with HFD group. This finding corresponds with the reported studies in literatures. Ghasemi [11] reported that Vitamin C co-administrated with HFD reduces the visceral fat accumulation induced or caused by HFD. In the study by Mamikutty et al. [7], vitamin C supplementation in Wistar rats fed with high-fat diet led to a reduction in fat storage which suggested that vitamin C might limit lipid synthesis and enhance lipid metabolism, which invariably might lead to the decrease in the abdominal circumference.

Visceral fat is linked with metabolic disorders, including insulin resistance and cardiovascular diseases. Vitamin C supplementation appeared to attenuate the increase in abdominal circumference as reported in the Vitamin C groups. The reduction in visceral fat may be attributed to vitamin C's role in reducing oxidative stress

and inflammation in adipose tissues, thereby inhibiting excessive fat storage.

Also, from this study, although there was an increase in body weight and abdominal circumference in the HFD group, there was no significant difference in the body length. There are limited studies on the effect of HFD and vitamin c on the body length in rats. However, Bergheim [12] reported that the metabolic stress induced by high fructose fed to male Wistar rats may interfere with overall growth or skeletal development and that chronic fructose consumption was also associated with disruptions in growth hormone signaling and bone health, potentially contributing to reduce body length. However, these changes could not be established in this study. This might be de due to the duration of the study or the dosage of the administered vitamin C.

The finding of this study also showed a significant increase in the BMI of the HFD group when compared with vitamin C treated groups. Vitamin C is an essential micronutrient and has many important biological functions, the most important of which is as an antioxidant and cofactor. In a study conducted by Ghasemi *et al.*, [11], male Wistar rats on a high-fructose diet showed an increase in BMI, which is used as an indicator of overall body adiposity.

The findings from this study revealed that there was no significant difference in the hematological variables and indices that were estimated and calculated, when the HFD group was compared with the control and vitamin C treated groups. This finding was however at variance with the study of Farah et al. [8] who reported that Wistar rats fed with high-fructose diet exhibited significant oxidative damage to erythrocytes, leading to decreased RBC counts. Additionally, Lanaspa et al. [13] found that vitamin C supplementation reduced markers of inflammation, including elevated WBC count and increased neutrophil-to-lymphocyte ratio, commonly induced by fructose. The reduction in WBC count reflects a decrease in systemic inflammation and improved immune function.

Platelet function and fibrinogen levels are critical for coagulation and hemostasis, both of which can be disrupted by oxidative stress and inflammation. In a study by Klein et al. [14], high fructose intake was associated with hypercoagulability, evidenced by elevated platelet counts and fibrinogen levels. When vitamin C was co-administered, it normalized these parameters, reducing the risk of thrombotic complications. High fructose levels can lead to erythrocyte membrane instability and increased susceptibility to hemolysis. Research by Rippe et al. [15], showed that vitamin C supplementation preserved erythrocyte morphology by preventing lipid peroxidation and protein oxidation in the cell membrane. This protective effect is vital for maintaining normal MCV and MCHC.

5. CONCLUSION

In conclusion, it has been established in this study that high fructose drink can alter hematological variable and indices through increased body weight and abdominal circumference. The mechanism of this effect maybe through lipogenesis and adiposity activation. Based on the findings of this study, it is therefore pertinent to recommend a prolonged study with high fructose drink and vitamin C intervention, in order to establish hematological alterations if any.

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